Getting to DIAG

How much does a journey really cost?

Explore (1) travel time, (2) money and (3) reliability.

Tell your story, then compare it with three other students.

Individual submission • Up to 3 points • **Synthetic data** will be provided for each student • Deadline: November 2nd, 2026

Your challenge

- Analyze your 20 **synthetic** one-way journeys to DIAG on 20 distinct weekdays
- Analyze the four students (you and the other 3) travelling on the same dates
- Design tables & graphs for individual (you) and group (all) analysis
- Compare your data with the group data
- Explain the goal of each table/graph

You can use AI for rendering the table or the graph but:

- You have to make this clear in the report, explaining how you interacted with the LLM
- The **design** of the table/graph **MUST** be done by you

The dataset: one row = one journey

Field	Type / unit	Meaning
student_id	Categorical	You + three fictional students
date	Temporal	Same 20 weekdays for all students
mode	Categorical	Walk, bicycle, public transport, car…
distance_km	Quantitative / km	One-way distance travelled
duration_min	Quantitative / min	Door-to-door time, including waiting
cost_eur	Quantitative / €	Direct cost of this one-way journey
arrival_margin_min	Quantitative / min	Required arrival time − actual arrival time

Positive margin = early; negative margin = late. Example: required 09:00, actual 09:08 → −8 min.

The dataset: example

student_id	date	mode	distance_km	duration_mi	cost_eur	arrival_margin_min
S1	10/5/26	Public transport	3.69	28	2.16	3
S2	10/5/26	Public transport	7.88	38	1.89	1
S3	10/5/26	Car	7.13	23	2.97	1
S4	10/5/26	Public transport	6.67	37	1.94	0
S1	10/6/26	Walk	3.27	48	0	5
S2	10/6/26	Public transport	7.85	44	1.89	-4
S3	10/6/26	Car	7.15	24	2.97	2
S4	10/6/26	Public transport	6.67	36	1.94	-2

You are the student S1. S2, S3, and S4 are the other 3 students.

Positive margin = early; negative margin = late. Example: required 09:00, actual 09:08 → −8 min.

Part A — Tell your individual story

- Create (at least) two views using your 20 journeys (minimum 1 table and 1 graph)
- Show temporal variation and local deviations from your usual travel time
- Compare modes using journey counts, typical duration and variability
- Report total direct cost, duration and the proportion of late arrivals
- Identify one practical trade-off: speed, cost or punctuality

Part B — Tell the group's story

- Create (at least) two additional views using all 80 journeys (minimum 1 table and 1 graph)
- Compare each student with the group; keep the individual visible
- Compare transport modes and inspect differences between students

Define the summaries before plotting

Describe the journey

Median/mean duration: typical journey

Range or quartiles: variability

Total cost: sum over the 20 journeys

Late rate: late journeys / all journeys

Make the group comparison explicit

Pooled median/average: all 80 journeys

Do not average individual averages

Late rate: count late journeys first and show the magnitude of the delay

Discuss distance and route context before interpreting differences between students

Overall or within each student?

Start with the pooled view

“Car journeys look faster.”

Which students mostly use the car?

Do they travel shorter distances?

How many journeys use each mode?

Then inspect the context

Compare modes within each student.

Display distance alongside duration.

Check whether the pattern persists.

Report limitations of the comparison.

Part C — Expectations and discoveries

- Before plotting the group data, write 3 testable expectations

  e.g. I expect car journeys to be faster…
- For each expectation, report: supported, contradicted or inconclusive
- Explore the same data and report 3 additional unexpected findings
- For every claim, cite a view and a numerical comparison
- If nothing surprises you, report the closest exploratory findings honestly

What counts as evidence?

Expectation → test

"I expect car journeys to be faster."

Compare average duration

Inspect the same comparison within students

Finding → bounded claim

Identify a visible pattern

Quantify it from your dataset

Consider a competing explanation

State what the data cannot establish

Tools and visual quality

- Use any tool: Excel + PowerPoint, Python, R, a visualization tool, or hand drawing
- Provide at least four views: two individual and two group views
- Give every view a readable title, labels, units and a legend when needed
- Use comparable scales for comparisons; start length-encoding bars at zero
- Show observations or variability when an average conceals the story
- A screenshot alone is insufficient: explain the design and the evidence

What to submit

- One short PDF report
- Include the four (or a few more) views and their task / data / encoding explanations
- Include three tested expectations and three exploratory findings
- Submit individually, even if you discuss ideas with classmates

Assessment — up to 3 points

Graph / Table design: 1.5 points

0.5 Coherent data and clear definitions

1.0 Task-driven views, readable encodings, context and variability

Analysis + explanation: 1.5 points

1.0 Claims supported by visual and numerical evidence; limitations

0.5 Task / data / encoding classification and clear report